Course Name:	Intelligent Mobile Robots
Course Number:	0135343
Project Name:	Autonomous Warehouse Robot: Full Navigation Loop
Project Grade:	**30 Marks**

1. Project Description and Objectives

Students build one simulated robot that does the whole job: it senses, localizes itself with a Kalman filter, maps the warehouse from LiDAR, plans a path with A*, follows it with Pure Pursuit, and replans when a box drops in its way.

Objectives:

- Integrate the methods of the six lab experiments into one working autonomy pipeline.
- Measure the pipeline's performance: pose error, path length against the optimum, and cross-track error.
- Analyse how errors in one stage affect the next.

Expected result: a robot that reaches its goal and replans around a new obstacle, with measured performance, a video and a short report.

2. Project Requirements

- Individual work.
- Use the project starter code (V2_Project_Student_Release.zip in the course repository): complete the TODOs in the four stage modules and in project_main.py.
- Each stage module must pass its own self-test before the stages are joined.
- Runs in the shared zuj-robotics environment (lab PCs) or the same environment on your own computer.
- AI assistants may be used for debugging and editing with a short disclosure; you must be able to explain any line of your code.
- All deliverables are submitted on Moodle by their phase deadline; late work loses 10% per day, up to three days.

3. Project Design Requirements

- Stage 2, localization: Kalman predict and update; tune Q and R so the filter beats raw GPS.
- Stage 3, mapping: world-to-grid conversion, LiDAR scan update and obstacle inflation.
- Stage 4, planning: an admissible heuristic and the A* loop on the inflated grid.

- Stage 5, tracking: the look-ahead point and the Pure Pursuit steering law, driven by the estimated pose.
- Integration: the full loop in project_main.py, including detecting the dropped obstacle and replanning.
- Performance is measured by grade_project.py, averaged over 5 fixed random seeds.

4. Suggested Project Structure

1. Run the starter code and read the pipeline: sense, localize, map, plan, track, replan.
2. Complete and self-test the localization stage.
3. Complete and self-test the mapping stage.
4. Complete and self-test the planning and tracking stages.
5. Wire the stages together in project_main.py and add replanning.
6. Tune Q, R and the look-ahead distance; measure with grade_project.py.
7. Record the video and write the report, including one extension of your own.

5. Partial Deliverables

Phase 1 – Problem Definition and Planning: Localization and Mapping (week 9, 10 marks)

- Stage 2 (Kalman filter) and Stage 3 (occupancy grid) pass their self-tests (4 marks each), plus a half-page plan of how the stages will be joined (2 marks).
- Deliverables: the two stage modules and the plan, zipped on Moodle.

Phase 2 – Design and Implementation: Planning, Tracking and Integration (week 14, 10 marks)

- Stage 4 (A*) passes its self-test (4 marks), Stage 5 (Pure Pursuit) passes its self-test (3 marks), and project_main.py runs the full loop with replanning (3 marks).
- Deliverables: the completed code, zipped as YourName_StudentID_Project.zip.

Phase 3 – Testing, Evaluation and Final Submission (week 15, 10 marks)

- Measured performance from grade_project.py over 5 seeds: pose error, path length and cross-track error (5 marks); a 1–3 minute video of the run (2 marks); a 1–2 page report with your analysis and extension, and a live demo (3 marks).
- Deliverables: grading_output.txt, the video link and the report PDF.

Submission: electronic copies only on Moodle; reports as PDF; videos as an unlisted YouTube or Google Drive link.

6. Scope and Final Outcome

Individual project: a complete simulated navigation loop for the warehouse robot (localize, map, plan, track and replan), built on the course starter code. Complete when every stage passes its self-test, the full loop reaches the

goal and replans around the dropped obstacle, and the measured performance, video and report are submitted in week 15.